Sections 9–11 at every rank step

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Status. This is a source-to-consumer project reconstruction, not independent acceptance. It separates two analytic supplier contracts, F and D, from the algebraic, selection and representation joins proved here. Their separate reconstruction and review remain required. The universal group uses the explicit fixed-rank weighted presentation below. No final inverse theorem is used as a premise.

1 The family at an arbitrary rank

Fix a standard $s \geq 3$, $m = s - 1$ and $1 \leq R \leq m$, and an unbounded limit integer N. All groups, lattices, metrics and finite dimensions below are standard unless a natural ultrapower is displayed or understood. Braces use $(-1/2, 1/2]$ throughout this packet. The Section 12 constructor allows either fixed half-open fundamental cube; use this same choice there. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be dense and internal. The input consists of a bounded function f on $[N]$, a global unit-vector multidegree $(1, m)$ nilcharacter $A(h, n)$, and an internal family of unit-vector degree–rank (m, R) nilcharacters χ_h. The latter have a fixed finite list of rational models and standard uniform Lipschitz bounds. For every $h \in H$, $\Delta_h f$ has uniformly positive degree-at-most-$m-1$ correlation with $A(h, \cdot) \otimes \chi_h$, with bounded internal witnessing tests. Every vertical identity is on the full natural extension of its group and compact quotient, not merely their standard points.

This is exactly the new input in rank-iteration001, Section 3. In particular no core/petal representation or independence statement is inherited from the preceding rank. They will be reconstructed. The corrected LP03 gives, after any dense internal restriction of H, many additive quadruples $h_1 + h_2 = h_3 + h_4$ for which

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)} \tag{1}$$

is $(m-1)$-biased, with standard uniform witnesses. This hereditary statement means reapplying LP03 to the restricted correlation hypothesis. It does not assert that a previously chosen dense set of quadruples survives restriction.

Analytic contract F. For internal functions $X_1, \ldots, X_4 : H \to {}^*\mathbb{T}$ and $\epsilon = N^{-j}$, if $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$ on many additive quadruples, there are a dense internal H', a standard finite K, and internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, $\delta \in {}^*\mathbb{T}$ such that

$$X_4(h) = \sum_{k=1}^{K} \{\alpha_k h\} \beta_k + \delta + O(\epsilon) \quad (h \in H').$$

The errors have a common standard bound. A permutation of the four positions changes which function is recovered. This is the precise Appendix F.1 input; the signed-graph and no-wrap argument is a separate analytic supplier.

Analytic contract D. For a polynomial sequence in one variable in a fixed rational filtered nilmanifold there is a factorization $g = \varepsilon g'\gamma$ with ε bounded, Lipschitz $O(1/N)$ and positive-index Taylor coefficients $O(N^{-i})$; g' takes values in a connected rational subgroup Q and is totally equidistributed on every standard finite cover of $Q/(Q \cap \Gamma)$; and γ takes values in a fixed discrete overgroup $\widehat{\Gamma} \supseteq \Gamma$ and is periodic modulo Γ with standard period. The subgroup Q is allowed to equal G. All factors remain adapted to the ordinary degree filtration. Only this one-variable contract is needed. Its quantitative published supplier and corrected proof are separate.

Theorem 1. *Subject to these two analytic contracts and the stated LP03 input, a dense internal subfamily has the following representation. There are fixed standard dimension blocks C, L, B, fixed filtered lattice-compatible maps from a weighted universal group U, and one integral top character η on $U_{(m,R)}$. Up to a fixed coordinate isometry ι,*

$$\iota\chi_h(n) = F_h(g_h(n)\Gamma, g_{0,h}(n)\Gamma_0) \qquad (n \in [N]).$$

The lower model has degree–rank at most $(m, R-1)$. The functions F_h are unit vectors with uniform standard bounds and exact top-character law; they need not be equal as functions. The horizontal coefficients are fixed core frequencies, linear petals $\{\alpha_a h\}\beta_a$ for $a \in L$, and independent petals for $a \in B$. The linear petals are Freiman homomorphisms on the retained set. For almost every additive quadruple, the core, all four independent-petal blocks and any three linear-petal blocks are independent modulo $O(N^{-i})$ separately at each weight i.

For every weight-m, length-R iterated commutator w, the real number $\eta(w)$ is zero if w contains two petals, or contains one letter from B. These are precisely the annihilations required by LP04. They hold for every new rank family satisfying the initial hypotheses, without inherited frequency information. The construction of the final two-variable test is the separate Section 12 join.

2 Selection and the corrected universal representation

We use the following elementary internal-choice rule, which also makes the countability uses precise. For an increasing internal sequence of witness sets S_k, suppose each member of an internal parameter set has a witness in some standard S_k. The least successful index is an internal integer. It is uniformly bounded: otherwise the internal nonempty sets on which it exceeds $1, 2, \dots$ have a common element by countable saturation, contrary to pointwise standard boundedness. At representative indices choose the first witness in a fixed choice order. This gives internal choices with one standard bound. Countable saturation here follows by diagonalizing nested finite intersections: at representative index v, choose a witness for the largest $k \le v$ whose first k conditions hold there. This k tends to infinity along the ultrafilter. The unsupported diagonal $x_{v,v}$ alone would not establish this conclusion.

We also need the "almost all" form. For nested internal bad sets B_k in a hyperfinite set X, if every B_k has infinitesimal relative size, the same diagonal construction chooses an unbounded K for which B_K has infinitesimal relative size and contains every standard B_k. Consequently, if their countable union is not contained in a sparse internal set, some B_k is dense. Applying finite pigeonhole inside that B_k fixes an integer coefficient vector and a standard error bound. This is the exact quantifier step used for failed frequency independence below.

First restrict to one of the fixed models for χ_h. Bounded tautification and normalization use their separately proved project interfaces. In a fixed model, only finitely many integral vertical characters can occur under a common Lipschitz bound. Indeed, along a fixed unit-speed top-torus direction X, the identity $F(\exp(tX)x) = \mathrm{e}(t\eta(X))F(x)$ and $\|F(x)\| = 1$ give $2\pi|\eta(X)| \le C_X \operatorname{Lip}(F)$ by taking

small t. On an integral basis this bounds all integer character coordinates. Finite pigeonhole therefore fixes the character.

For each weight i, choose finitely many elements of $\Gamma \cap G_{(i,0)}$ whose logarithms generate its rational Lie algebra. Such elements are obtained by clearing denominators in an adapted rational Mal'cev basis; their real generated subgroup is $G_{(i,0)}$. Use one formal letter of weight i for each. Define U by killing all Lie words of weight greater than m and all weight-m words of length greater than R. Set

$$U_{(d,r)} = \exp \operatorname{span}\{\text{surviving words of weight} > d, \text{ or weight} = d \text{ and length} \geq \max(r, 1)\}$$

for positive d, with $U_{(0,0)} = U$. This is the exact LP04 weighted presentation before the additional petal relations. Its Hall-coordinate lemma supplies the lattice generated by the letters and its filtered intersections. The upward-closed word conditions prove every bracket inclusion. In particular $U_{(d,0)} = U_{(d,1)}$.

Mapping each letter to its chosen lattice element gives a filtered homomorphism $\phi : U \to G$. The killed words vanish by the degree–rank bracket law in G; the map sends the generated lattice into Γ. It is onto each $G_{(i,0)}$. In $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$ the images of the weight-i letters span the real vector space and generate a full sublattice. Thus the induced map of compact horizontal tori is onto. Select internal preimages of $\mathrm{Taylor}_i(g_h\Gamma)$ to get frequencies $\xi_{h,i,a} \in {}^*\mathbb{T}$. Pull the fixed top character back by ϕ and initially take G_0 trivial. This proves the exact Definition 9.11 representation, including lattice compatibility and all-limit equivariance, for every new family.

Here and below horizontal Taylor coefficients are homomorphisms on the polynomial-sequence group, are unchanged by translating n, and for $c \in G_{(i,1)}$ the sequence $c^{\binom{n}{i}}$ has only the ith horizontal coefficient c and zero others. These facts use the iterated-difference commutator rule B.7: all noncommutative correction terms have rank at least two and disappear modulo $G_{(i,2)}$; an extra difference has weight above i. Binomial differences give the last assertion. Polynomial products use B.4. No claim that arbitrary Taylor factors commute is used.

If $\xi_{i,a} = \sum_b c_{i,a,b}\xi'_{i,b}$ with standard integers $c_{i,a,b}$, map the new weight-i letter $e'_{i,b}$ to $\prod_a e_{i,a}^{c_{i,a,b}}$ in a fixed order. The universal relations and lattice inclusion hold, giving $\psi : U' \to U$. In the abelian horizontal quotient this is exactly the indicated integer matrix. Replace ϕ by $\phi\psi$ and η by $\eta\psi$; the original orbit and test are unchanged. When these matrices depend on h, the preceding bounded-choice rule and finite pigeonhole fix them on a dense internal subfamily before this step.

3 Frequency regularization with the actual quantifiers

Fix one weight and put $\epsilon = N^{-i}$. A partial description consists of fixed core frequencies, independent-candidate petals b_h, bracket-linear petals l_h, fixed rational frequencies, and $O(\epsilon)$ errors, with all original frequencies standard integer combinations of them. Every list is standard finite, and every family is internal. Dense restriction fixes the integer matrices and error bounds. Put all original frequencies in b_h to obtain an initial partial description.

Minimize lexicographically the numbers of b petals, then l petals, then core frequencies, among all such partial descriptions on dense subsets. Bracket-linear petals are required to be individually Freiman homomorphisms. This costs only a finite dense restriction: place each $\{\alpha h\}$ in one of a fixed finite collection of intervals of width less than $1/4$. For an additive quadruple the difference of the four fractional parts is an integer of absolute value less than one, hence zero. Multiplication by any real β preserves that identity. The intervals may be anywhere in the fundamental cube; none is required to surround zero.

If independence of the core, four b blocks and the first three l blocks fails on more than a sparse set of quadruples, the bad-set rule above fixes one nonzero standard integer relation with error $O(\epsilon)$ on many quadruples. There are three exhaustive cases.

If it contains only core coefficients, solve for a core with nonzero coefficient a. Choose internal ath roots of the other core frequencies. Division introduces a residue in the finite group $a^{-1}\mathbb{Z}/\mathbb{Z}$ and an $O(\epsilon)$ error. Move them to the rational and error lists. This reduces the core count without changing either petal count.

If the relation involves l but no b, it is a relation on three free parameters. Fix two by averaging over the many triples. On a dense set of the remaining parameter, solve for one l coefficient. Divide the remaining bracket-linear coefficients as real numbers β/a, and root the core coefficients. Freeze the finite rational carry and add the values at the two fixed parameters to the core. A further fixed-box restriction restores the Freiman laws for the divided bracket-linear terms. This reduces the l count without changing the b count.

If a b coefficient occurs, use a symmetry of $h_1 + h_2 = h_3 + h_4$ to place a nonzero one in position four. The Freiman laws allow elimination of the fourth l block after this permutation. Group the relation as $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$, with X_4 containing that b coefficient and fixed core terms. Contract F expresses X_4 as finitely many bracket-linear terms plus a constant and error. Solve for the chosen b coefficient, using roots of the other b and core frequencies and dividing the real bracket-linear coefficients. Freeze the finite rational carry, enlarge the other lists, and refine into fixed boxes as above. The original frequencies are still integer combinations of the new lists, because the old frequencies are integer multiples of their chosen roots. The b count has decreased. Each case contradicts the lexicographic minimum.

Thus the required independence holds for almost all quadruples. Apply this argument successively for the finitely many weights $1, \ldots, m$. Dense restrictions preserve previously obtained independence: their bad sets remain $o(N^3)$, while the retained parameter set has standard positive density. Standard integer relations, rational carries, error constants, finite dimensions and fixed model choices are selected internally with the explicit bounded-choice rule. This proves the regularization step of 10.5–10.7 conditional only on contract F, with its needed family uniformity.

4 Removing rational and small coordinates without losing unit norm

Apply the integer change-of-basis construction to the regularized frequency lists. We now remove each rational or small coordinate in turn. The remaining top character and lattice map must be updated; forgetting them would not give a new representation.

For a small coordinate $\xi = O(N^{-i})$, choose its small real lift and put $c = \phi(e_{i,a})^{\xi} \in G_{(i,1)}$. Set $g'_h(n) = c^{-\binom{n}{i}} g_h(n)$. B.4 proves polynomiality and horizontal homomorphism shows that exactly the selected coordinate has been removed. On $[N]$, $c^{\binom{n}{i}}$ is a bounded smooth function of $u = n/(2N)$. Indeed its logarithm is $\binom{2Nu}{i}\xi \log \phi(e_{i,a})$; its coefficients and all derivatives on a fixed compact interval are bounded because $N^i\xi = O(1)$. Extend this logarithm to the unit circle by using its real formula on the lift $(-1/4, 3/4)$ and multiplying by a fixed smooth cutoff equal to one on $[0, 1/2]$ and zero near the lift's endpoints. Let $c_h(u)$ be its exponential. Replace the test by

$$F'_h(x, x_0, u) = F_h(c_h(u)x, x_0).$$

It is globally a unit vector and uniformly Lipschitz. The top image $\phi(U_{(m,R)})$ is central, so the same exact vertical law holds everywhere. Add the circle orbit $n/(2N)$ to the lower model. It has

degree one, which is at most $(m, R-1)$ because $m \geq 2$. Thus this is an exact representation, not an approximation to the original sequence.

For a rational coordinate $q \in \mathbb{Q}/\mathbb{Z}$, put $\gamma = \phi(e_{i,a})^q$ and $g'_h(n) = g_h(n)\gamma^{-\binom{n}{i}}$. Again only that horizontal coordinate is removed. The element γ is rational relative to Γ. The subgroup $\widehat{\Gamma} = \langle \Gamma, \gamma \rangle$ is discrete: in rational Mal'cev coordinates the finite nilpotent collection formulas express every word as a polynomial in integer exponents of a finite set of generators and their finitely many commutators; clearing their fixed rational denominators gives a discrete coordinate grid. Since Γ is cocompact, its index in $\widehat{\Gamma}$ is finite. Its normal core $\Gamma' = \bigcap_{\delta \in \widehat{\Gamma}} \delta \Gamma \delta^{-1}$ is therefore a finite-index subgroup, normalized by γ.

Choose a standard b with $\gamma^b \in \Gamma$ and a standard period q_0 such that $\binom{n+q_0}{i} \equiv \binom{n}{i} \pmod{b}$. For example a sufficiently divisible multiple of $b\,i!$ works, by the integer binomial difference formula. Put $\gamma_r = \gamma^{\binom{r}{i}}$ for $r \pmod{q_0}$. Then

$$F_h(g_h(n)\Gamma, x_0) = F_h(g'_h(n)\gamma_r\Gamma, x_0) \quad (n \equiv r \pmod{q_0}).$$

The test $W_{r,h}(x\Gamma', x_0) = F_h(x\gamma_r\Gamma, x_0)$ is well-defined, unit, and has the same top-frequency law. Also $W_{0,h}(x\Gamma', x_0) = F_h(x\Gamma, x_0)$ is such a global unit test.

To interpolate the finitely many residue tests, choose disjoint small arcs around r/q_0 in $\mathbb{T}$. In the rth arc choose a smooth angle θ_r which is zero at r/q_0 and $\pi/2$ near its boundary. Set $a_r = \cos\theta_r$ there and zero outside; set $b = \sin\theta_r$ there and $b = 1$ outside all arcs. The supports of the a_r are disjoint and $\sum_r a_r^2 + b^2 = 1$. The test

$$\left(\sum_r a_r(u) W_{r,h}(x, x_0),\ b(u) W_{0,h}(x, x_0) \right)$$

is globally Lipschitz, unit and has the same vertical character. At the orbit $u = n/q_0$ it is exactly $(\chi_h(n), 0)$. This fixed zero padding is explicit; we do not assert that an arbitrary scalar interpolation stays unit or that the old dimension has been retained. Correlation and subsequent matrix comparisons recover the original family by the fixed coordinate projection.

The map from the remaining universal group must now land in the covering lattice Γ'. Choose a standard M for which every remaining lattice generator's Mth power lies in Γ'. Send each new letter to that Mth power, and divide its horizontal frequency by M. Pull back the character by the same map. For linear petals choose the division as $\{\alpha h\}(\beta/M)$; for core coordinates choose fixed roots and for independent coordinates choose internal roots. Repeating this finite process produces a common standard denominator. It is not silently dropped, as in the displayed conclusion of the printed proof of 10.10.

Independence is preserved: multiply an alleged standard integer relation between divided frequencies by the common denominator to contradict the old independence. After dividing linear petals, refine again into the fixed short boxes to make their real fractional-part sums exactly additive, hence their divided values Freiman. This is valid at arbitrary box locations. All rational covers, maps, dimensions, and periods are fixed on a dense internal subfamily before use. Small-coordinate curve bounds are uniform by the internal bounded-choice rule. A finite composition of paddings is one fixed coordinate isometry. We have proved the precise representation part of Theorem 1, without replacing the varying F_h by a single equal function through an approximation argument.

5 The horizontal consequence of factorization, with denominators

Here we derive the needed D.6 conclusion from contract D. View a degree–rank group as ordinarily filtered by $G_i = G_{(i,0)}$. We retain the finer horizontal spaces $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$. For a polynomial orbit $g\Gamma$ put

$$\Xi_i = \{\lambda \in \widehat{\mathrm{Horiz}_i(G)/\Gamma} : \lambda(\mathrm{Taylor}_i(g\Gamma)) = O(N^{-i}) \pmod 1\}.$$

Characters here have standard integer coordinates. Their real linear lifts define $\Xi_i^{\perp}$ inside the real horizontal vector space.

Suppose F is a bounded Lipschitz test with nonzero standard average. Take one scalar component if needed, and factor $g = \varepsilon g'\gamma$. On a progression of the standard periodic modulus, choose a bounded standard representative $\gamma_P \in \widehat{\Gamma}$ of the constant coset of γ. Set $Q_P = \gamma_P^{-1} Q \gamma_P$ and $g_P = \gamma_P^{-1} g' \gamma_P$. The finite-cover equidistribution clause in D, applied to the common finite-index lattice, makes g_P totally equidistributed in $Q_P/(Q_P \cap \Gamma)$ on every retained dense progression. Rationality of γ_P gives rationality of Q_P.

Partition into those residue progressions and then into intervals of length δN for a sufficiently small standard δ. Averaging selects a piece with an average bounded below by the original fixed threshold. On that piece freeze $\varepsilon(n)\gamma_P$ at one point. The uniform Lipschitz error is $O(\delta)$, so equidistribution gives

$$\left| \int_{P/(P\cap\Gamma)} F(\varepsilon_0 x)\, d\mu(x) \right| > c > 0 \tag{2}$$

for the connected rational $P = Q_P$ and bounded ε_0. This also holds for an internal test with a standard Lipschitz bound: compactness gives a finite standard uniform net in that Lipschitz ball. Apply total equidistribution to each net entry and then let the fixed net error tend to zero. Thus no unjustified standardness of the selected F_h or of the frozen bounded translate is being used.

We claim $\pi_i(P \cap G_i) \supseteq \Xi_i^{\perp}$. Let λ be an integer character annihilating the rational space $\pi_i(P \cap G_i)$. Then $\lambda(\mathrm{Taylor}_i(g_P)) = 0$. Conjugation changes no horizontal coefficient since $[G, G_i] \subseteq G_{(i,2)}$. Hence the horizontal homomorphism identity gives

$$\lambda(\mathrm{Taylor}_i(g)) = \lambda(\mathrm{Taylor}_i(\varepsilon)) + \lambda(\mathrm{Taylor}_i(\gamma)) \pmod 1.$$

The first term is $O(N^{-i})$. The second is rational of bounded standard denominator: derivatives of γ lie in $\widehat{\Gamma} \cap G_i$, whose horizontal image is a finite-index over-lattice of that of $\Gamma \cap G_i$. Thus for some standard positive q, $q\lambda \in \Xi_i$. Every annihilator of the rational image has this property. Taking real annihilators proves the claim. In particular no false assertion that γ takes values in Γ or has zero horizontal coefficient is needed.

6 Applying the coset conclusion to the actual quadruple

After all the preceding dense restrictions, reapply hereditary LP03. There are $\gg N^3$ biased quadruples. The exceptional sets for frequency independence have size $o(N^3)$. Permutations preserving the additive quadruple equation make the omitted linear-petal position any of the four; the union of the four exceptional sets is still sparse. Fix a biased quadruple with all four required independence statements.

Use the representations of its four nilcharacters and the lower bias test to form a product group

$$\mathcal{G} = G^{(1)} \times G^{(2)} \times G^{(3)} \times G^{(4)} \times G^{(0)}.$$

The first four groups have degree–rank at most (m, R); the last has degree at most $(m, R-1)$, including all four lower representation models and the ordinary degree-at-most-$m-1$ bias witness. The stronger strict inequality printed for this last group is unnecessary and is not assumed. Translations in n preserve horizontal Taylor coefficients by their homomorphism/difference identity. Zero extensions in (1) introduce only an interval-intersection mask. Approximate this mask in normalized L^1 to error δ by a bounded Lipschitz circle test in $n/(cN)$, with fixed large c and Lipschitz constant $O(1/\delta)$. Its two transition strips have total length $O(\delta N)$; multiplying by the uniformly bounded remaining tests controls the average error. Choosing δ below the fixed bias threshold preserves positive bias. This circle has degree one and belongs in $G^{(0)}$ because $m \geq 2$.

The resulting product test has a nonzero scalar average and exact vertical law

$$F(\phi_1(t_1)x_1, \dots, \phi_4(t_4)x_4, x_0) = \mathrm{e}(\eta(t_1) + \eta(t_2) - \eta(t_3) - \eta(t_4))F(x_1, \dots, x_4, x_0).$$

All factors, including the cutoff, are uniformly Lipschitz on a fixed standard product geometry. Apply the preceding factorization consequence to obtain a connected rational $P \subseteq \mathcal{G}$ and the nonzero coset integral (2).

7 Rational duality and the two commutator conclusions

For each weight i, let $V_{123,i}$ be the real space spanned by the three-coordinate diagonal images of core letters and by each individual coordinate image of every petal letter in positions $1, 2, 3$. Define $V_{124,i}$ likewise. The projection of $\pi_i(P \cap \mathcal{G}_i)$ to these three factors contains the indicated V.

Here is the exact duality argument. If containment failed, rationality gives a rational linear functional that vanishes on the projected space but not on V. Extend it by zero in the omitted factors and clear denominators to make it an integer character λ. Because the full space contains $\Xi_i^{\perp}$, this character is in the rational span of Ξ_i. After multiplication by another standard integer it belongs to Ξ_i itself. Evaluating it on the Taylor coefficient gives a relation

$$\sum_{a \in C} (c_{1a} + c_{2a} + c_{3a})\xi_{C,a} + \sum_{k=1}^{3} \sum_{a \in L \cup B} c_{ka}\xi_{h_k,a} = O(N^{-i}).$$

The selected independence forces every petal coefficient and each displayed core sum to vanish. That says the character vanishes on V, a contradiction. The 124 proof is identical. This uses rational real subspaces, not an unjustified identification of an arbitrary real annihilator with one integral member of Ξ_i.

Similarly the projection onto all four main factors contains the space $V_{B,i}$ generated by diagonal core images and by the individual-coordinate images of every letter in B. In the analogous integer relation eliminate the fourth L coefficient using $\xi_{h_4,a} = \xi_{h_1,a} + \xi_{h_2,a} - \xi_{h_3,a}$ for $a \in L$. Independence makes the four B coefficients vanish separately and the sum of core coefficients vanish. This is exactly annihilation of $V_{B,i}$. No numerical range for B is borrowed from the conflicting printed order of the L and B blocks.

Let w be any bracketing of R letters of weights summing to m. First suppose it has two petal occurrences. For one occurrence lift its first-coordinate image, with zero second and third coordinates, from $V_{123,i}$ to $P \cap \mathcal{G}_i$. For the second lift its first-coordinate image with zero second and fourth coordinates from $V_{124,i}$. Lift the remaining letters so that their first horizontal coordinate is the required one. Take the same iterated commutator of these lifts.

In the first factor it equals $\phi_1(w)$ exactly. Replacing one lift by its horizontal error inserts an element of $G_{(i,2)}$ in place of $G_{(i,1)}$, raising the total rank to at least $R+1$ at weight m; every such

term vanishes. Higher BCH terms have larger weight and vanish as well. In factor two, either selected petal lift has such an error; factor three has the first, and factor four the second. Therefore those coordinates are the identity. In factor zero, an R-fold commutator of the prescribed weights lies in its trivial (m, R) group. We obtain

$$(\phi_1(w), 1, 1, 1, 1) \in P. \tag{*}$$

This reasoning applies to each position of a repeated letter and to every bracketing, not only a single left-nested commutator.

If w has a B letter, use $V_{B,i}$ to lift that occurrence with all other main coordinates zero horizontally. Lift the remaining occurrences by their first coordinates and repeat the same rank-error calculation. This proves (*) as well. It works for $R = 1$; the other top coordinates then already vanish, and the lower model has trivial weight-m group.

Finally (*) gives a real-zero conclusion. The element is central in $\mathcal{G}$. Since P is a connected subgroup of a simply connected nilpotent group, its entire one-parameter subgroup belongs to P. Left translation by its tth power leaves Haar measure on $P/(P \cap \Gamma)$ invariant and commutes with the bounded coset translate. The vertical law multiplies the nonzero integral by $\mathrm{e}(t\eta(w))$. Hence $\mathrm{e}(t\eta(w)) = 1$ for every real t, so $\eta(w) = 0$ as a real number. Using only $t = 1$ would give an integer condition and would be insufficient. This proves both annihilations in Theorem 1.

8 The LP04 and rank-family joins

The output has the exact LP04 hypotheses: the corrected universal weighted group, an integral top character with all required real-zero annihilations, fixed lattice-compatible ϕ, matched horizontal Taylor tori, and a unit representation with a lower degree–rank factor. Its petal quotient kills the whole ideal generated by the bad words at every weight; the vanishing just proved concerns the top restriction needed to descend the character. These distinct roles are not identified.

The Section 12 join chooses its stronger integral extension to the ordinary top before invoking LP04 comparison, and gives one global multidegree $(1, m)$ nilcharacter equal to a fixed coordinate embedding of ζ_h. LP04 then supplies the whole matrix $(\iota\chi_h) \otimes \overline{\zeta_h}$ with strictly lower degree–rank. Projecting the first tensor factor through the fixed coordinate projection gives $\chi_h \otimes \overline{\zeta_h}$ with the same bound and exact values. Thus the unit padding used in rational removal does not weaken the actual matrix input required by rank-iteration001.

All selections, roots, covers, maps and tests were made internally on representatives; each retained dimension, character and bound is standard. There are finitely many weights and coordinates at any one rank, so all density and correlation losses remain positive standard numbers. The next rank family from rank-iteration001 again has exactly the hypotheses of Section 1. Repeating the present reconstruction at that new rank uses its own new frequency data. No independence is transported through a Stone–Weierstrass approximation or assumed for the newly selected test. There are m rank steps, so the same finite-loss conclusion holds across the full iteration. This is the required source-to-consumer compatibility, conditional on the two separately supplied analytic contracts and the independent acceptance of the project proof units used here. Representation equality is required on $[N]$. Whenever a global sequence is used, take the displayed polynomial model as its extension. Small-frequency removal need not preserve a previous extension outside $[N]$; the interval mask in the quadruple argument is precisely what makes that change harmless.

9 Bounded correction and consistency record

The source scope is final GTZ Sections 9–11, the immediate pp. 1307–1309 consumer, the D.4–D.6 interface, B.4/B.7, and the selection statements in Appendix A. The attached record pins exact passages and the current project consumers. Concrete changes are:

1. State the fixed rank R explicitly in Definition 9.1's second truncation relation, where the printed formula writes r without the fixed rank subscript. The printed weight cutoff $\sum i_j \geq s$ is already equivalent to $\sum i_j > m$ and is not an error. Rendering the source rather than relying on text extraction establishes this distinction.

2. Retain the common division from 10.9 in 10.10 and restore Freiman laws after real coefficient division. Fixed arbitrary-location boxes suffice.

3. Rational removal uses a finite cover and explicit same-character unit completion with a fixed isometry, rather than unproved unit interpolation.

4. Keep varying uniformly Lipschitz F_h; total boundedness does not make them literally equal as claimed in the immediate p. 1308 narrative.

5. Use named core/linear/independent blocks, because p. 1302 reverses the order used in the statement of 10.10 and Theorem 11.1.

6. Treat the repeated clauses (iii),(iv) of Theorem 11.1 as duplicates; there are two distinct annihilation conclusions.

7. Use degree–rank at most $(m, R-1)$ for the lower product model, not an unneeded strict inequality below that index.

8. Retain the rational Taylor term in D.5 and clear its denominator in the annihilator argument. The factorization subgroup may equal the ambient group. The separate analytic supplier addresses the D.2 total-equidistribution issue and F.1 signed-graph/no-wrap details.

9. The real-zero character argument uses all real powers in the connected subgroup. One integral phase identity alone is insufficient.

10. Reapply LP03 after each dense restriction; intersect many biased quadruples with almost-all independence, not two unrelated dense sets.

These are authored repairs and scoped dependencies, not a declaration that all historical versions or the entire inverse theorem have been checked. The original published source and all earlier frozen packets are preserved.